

Fractal-Guided Circuit Testing Algorithm for Efficient Mixed-Signal VLSI Verification and Validation

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Abstract:

Mixed-signal Very Large-Scale Integration (VLSI) circuits involving a combination of analog and digital components are encountering growing verification and validation problems because of complicated interactions, process variability and nonlinearities. The available methods of testing are time-consuming, they do not cover as many faults as possible, and they do not always analyze the most important components of the circuit in detail. The current study introduces a new Fractal-Guided Adaptive Resonance Testing (FGART) model that is the first model to combine the multi-scale fractal-based test vector generation with the adaptive resonance-based fault detection in mixed-signal VLSI verification. FGART quickly concentrates the testing on risky circuits areas and offers far superior coverage of faults compared to deterministic, random, or fractal-only approaches and uses less time and power. Multi-scale test vectors produced by FGART are based on fractal inspired metrics and it also adapts dynamically to the resonance patterns of the circuit by changing the test intensity. This is a good way of facilitating concentration of testing resources in regions that are high risk. This hybrid method will guarantee that the behaviour of analog and digital components are both exercised efficiently, with no extra simulations of duplicate responses, and subtle parametric and functional faults are targeted. Testing of FGART on standard mixed-signal benchmarks indicates a total test-time reduction of 40 percent and a 22 percent improvement in fault coverage compared to the classical random, deterministic, and fractal-only test cases. Moreover, FGART is highly scalable to larger and more complex VLSI designs whilst remaining efficient with an increase in the complexity of the circuit. To conclude, FGART is an innovative, intelligent, adaptive, framework of mixed-signal VLSI verification and validation; it is capable of being fast, reliable, and less abstraction among its computational...

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